

AI-Driven Profit Optimization in Agricultural Markets

A Transport Cost-Aware Decision Framework

Course: EAL421 – Application of AI and ML in Agriculture

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Introduction

Opening Hook:

- ▶ Imagine two markets offering 2000 and 2300
- ▶ Most people choose 2300
- ▶ But transport cost may reduce it to 1800

Key Idea: This project helps farmers choose the most profitable market.

Analogy: Like Swiggy — food price looks cheap, but delivery decides real cost.

Problem Statement

The Hidden Problem in Agri Business

- ▶ Same crop has different prices across mandis
- ▶ Farmers sell in nearest market
- ▶ Transport cost is ignored
- ▶ No integrated decision system

Core Statement: Farmers lack a system to maximize net profit, not just selling price.

Analogy: Choosing a job based only on salary without checking rent.

Methodology

System Workflow

- ▶ Collect mandi price data
- ▶ Calculate distance and transport cost
- ▶ Train ML model for price prediction
- ▶ Compute net profit
- ▶ Recommend best market

Formula:

$$\text{Profit} = \text{Predicted Price} - (\text{Distance} \times \text{Cost/km})$$

Analogy: Like Google Maps — shortest route is not always fastest.

Data Description

Data Inputs

- ▶ Crop type
- ▶ Market (mandi)
- ▶ Date
- ▶ Modal price
- ▶ Distance (km)
- ▶ Cost/km
- ▶ Arrival quantity

Example Dataset:

Market	Price	Distance	Cost/km	Profit
A	2000	10	5	1950
B	2300	50	5	2050
C	2100	20	5	2000

Insight: Higher price does not mean higher profit.

Analogy: Distance is money leaking from profit.

Model / Approach

Machine Learning Model

- ▶ Linear Regression
- ▶ Optional: Decision Tree

Steps

- ▶ Predict price
- ▶ Calculate transport cost
- ▶ Compute net profit
- ▶ Rank markets

Output: Best Market → Expected Profit 2050

Analogy: Normal decision = guess ML decision = data-driven choice

Results & Analysis

Key Findings

- ▶ Nearby markets are not always best
- ▶ Profit depends on price-cost balance
- ▶ Significant profit differences exist

Graphs to Include

- ▶ Bar chart: Profit comparison
- ▶ Line chart: Price trends
- ▶ Scatter: Distance vs profit

Case Insight: Tomato prices vary across mandis → affects profit.

Analogy: Cheap product + high delivery = regret.

Conclusion & Future Scope

Conclusion

- ▶ Converts data into decisions
- ▶ Improves farmer profitability
- ▶ Reduces guesswork

Future Scope

- ▶ Mobile app for farmers
- ▶ Voice assistant (local languages)
- ▶ Real-time mandi integration
- ▶ Weather + demand analysis

Closing Line: Farming is business, and business needs smart decisions.

Analogy: From instinct to intelligence.

Questions?

Prepared Answers

- ▶ Why ML? → Predict future prices
- ▶ Is it practical? → Uses real data
- ▶ Transport changes? → Can update dynamically

Final Note: This can become a real-world farmer decision app.